PrIdentity: Generalizable Privacy-Preserving Adversarial Perturbations for Anonymizing Facial Identity

Saheb Chhabra, Kartik Thakral, *Student Member, IEEE*, Richa Singh, *Fellow, IEEE*, and Mayank Vatsa, *Fellow, IEEE*

***Abstract*— With the rapid proliferation of face recognition systems, the risk of privacy leakage from facial images has become a pressing concern. Applications such as Find Face and Social Mapper can readily expose an individual's identity without consent. Existing anonymization approaches partially address the problem: synthesis and fusion-based methods suppress identity but often distort facial attributes, reducing data utility, while adversarial perturbation methods improve generalizability across recognition models but rely on fixed $\mathcal{L}_1$ or $\mathcal{L}_2$ norms, leading to underfitting or overfitting. As a result, no single method jointly satisfies the three essential properties of effective anonymization: privacy, data utility, and generalizability. To address these limitations, we present a novel *Privacy-Preserving Identity Anonymization (PrIdentity)* algorithm that anonymizes the identity of a given image while preserving privacy. Our approach learns adversarial perturbations through an $\mathcal{L}_p$ norm based regularization technique, maintaining a balance between privacy and data utility. Furthermore, we ensure the anonymized images generalize effectively across different unseen face recognition models. To the best of our knowledge, this is the first work to introduce a learnable p parameter in the $\mathcal{L}_p$ norm for privacy-preservation. We evaluate PrIdentity on the LFW, CelebA, and CelebA-HQ datasets across multiple face recognition architectures, complemented by a user study on both original and anonymized images. The results demonstrate that our algorithm effectively conceals identities while preserving visual appearance, achieving state-of-the-art performance in identity anonymization. We also carry out bounding box distance prediction experiments to validate data utility, attaining a state-of-the-art Euclidean distance of 2.65, which is 1.17 lower than the second-best method.**

***Index Terms*— Privacy-preservation, adversarial perturbations, face recognition, identity anonymization.**

Received 24 April 2025; revised 8 July 2025 and 17 September 2025; accepted 12 October 2025. Date of publication 27 October 2025; date of current version 22 January 2026. This work was supported by Srijan: Center of Excellence (COE) on GenAI through IndiaAI Mission and Meta. The work of Kartik Thakral was supported by the Prime Ministers Research Fellowhip. This article was recommended for publication by Associate Editor Z. Lei upon evaluation of the reviewers' comments. *(Corresponding author: Richa Singh.)*

Saheb Chhabra is with the Department of CSE, Indraprastha Institute of Information Technology Delhi, New Delhi 110020, India (e-mail: sahebc@iiitd.ac.in).

Kartik Thakral, Richa Singh, and Mayank Vatsa are with the Department of CSE, Indian Institute of Technology Jodhpur, Jodhpur 342037, India (e-mail: thakral.1@iitj.ac.in; richa@iitj.ac.in; mvatsa@iitj.ac.in).

Digital Object Identifier 10.1109/TBIOM.2025.3625986

I. Introduction

As PER Global Social Media Stats 2024 [1], the number of social media users from January 2023 to January 2024 increased by 320 million worldwide. However, with the rise in the number of social media users, identity theft [2], [3] and financial fraud [4], [5] via online social media profiling have also increased. Fig. 1 illustrates the information profiling via social media platforms. This leads to a decline in public trust in social media and presents a serious threat to an individual's privacy. To mitigate this risk, it is essential to perform **face de-identification**, often referred to in prior work as face anonymization, to minimize misuse caused by automatic identity leakage.

In this work, we formulate the requirements for effective face anonymization as satisfying three fundamental properties: 1) **Privacy** — suppressing identity cues so that automated face recognition systems fail to identify the individual, 2) Data Utility — ensuring minimal perceptual change so that images remain visually realistic and useful for downstream tasks, and 3) Generalizability — ensuring anonymization effectiveness across unseen face recognition (FR) models. While these properties are widely acknowledged as desirable, they are rarely addressed jointly in prior literature. Existing approaches predominantly target individual properties, often at the expense of others. For example, synthesis-based methods [6], [8], [11] strongly achieve privacy by generating new faces or fusing target identities, but frequently compromise data utility due to visible distortions or unrealistic artifacts. k-Anonymity-based methods [12], [13] offer theoretical privacy guarantees but typically result in significant visual degradation. Adversarial perturbation methods [14], [15] show promising generalizability by leveraging transferability across models, yet they rely on fixed $\mathcal{L}_1$ or $\mathcal{L}_2$ constraints, which lead to suboptimal performance: the $\mathcal{L}_1$ norm introduces localized, high-magnitude pixel changes that may leave identity cues untouched, while the $\mathcal{L}_2$ norm generates smoother perturbations that are often perceptible and insufficiently targeted.

This imbalance creates key technical challenges in designing an effective anonymization algorithm. It must generate perturbations that both maximize privacy and preserve visual utility, while avoiding overfitting to known face recognition models to ensure transferability to unseen ones. Additionally,

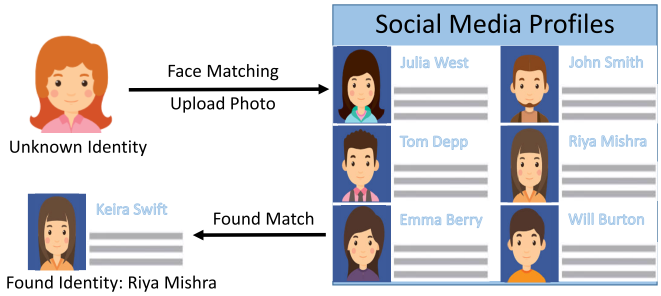

Fig. 1. Illustration of information profiling of unknown identity from social media platforms.

it should prevent degenerate perturbations, such as high-frequency noise, that would degrade image realism or generalization.

To address these challenges, we propose **Privacy-Preserving Identity Anonymization (PrIdentity)**, which learns adversarial perturbations to explicitly optimize the three properties. Our method maximizes the distance between the face representations of the original and perturbed images while preserving the visual appearance, using principled regularization inspired by Szegedy et al. [16] and Carlini and Wagner [17]. Instead of relying on fixed $\mathcal{L}_1$ or $\mathcal{L}_2$ norms, we learn the parameter p in the $\mathcal{L}_p$ norm, enabling the model to adaptively balance sparsity and smoothness. This avoids the pitfalls of overfitting or visible artifacts, and maintains a stable trade-off between privacy and data utility. Moreover, leveraging the well-established transferability of adversarial perturbations across deep models [18], [19], [20], PrIdentity generalizes effectively to unseen face recognition models without requiring any target identity image. We further extend our approach to anonymize multiple images via a single uniform perturbation, enabling scalable suppression of an identity across a user's multiple social media photos. We validate the efficacy of PrIdentity through extensive experiments on the LFW [21], CelebA [22], and CelebA-HQ [23] datasets, as well as a user study to confirm that the perturbations remain imperceptible to humans while deceiving FR networks. To summarize, our work addresses the limitations of existing methods by proposing a principled solution that jointly ensures privacy, data utility, and generalizability. The key research contributions of this work are:

- We propose PrIdentity, a novel face anonymization algorithm that learns the p parameter in the $\mathcal{L}_p$ norm, allowing adaptive control over perturbation sparsity and smoothness.
- We design a unified objective that jointly optimizes privacy and data utility using a margin-based loss function, explicitly balancing identity suppression and visual realism.
- Our method achieves robust black-box generalization, effectively anonymizing faces against unseen face recognition models without requiring target identity images.
- We introduce a single vs. multiple image anonymization use case, enabling the learning of a single perturbation applicable across multiple images of the same identity for scalable social media privacy-protection.

II. Related Work

In the face-based privacy-preserving literature, researchers have proposed various methods to suppress (i) attributes such as gender and age [24], [25], [26] or (ii) facial identity for privacy-preservation [8], [10], [27], [28]. Our focus in this research is identity anonymization. Existing surveys typically classify anonymization methods based on the stage of operation or data representation. For example, taxonomy proposed by Meden et al. [29] categorizes privacy-enhancing technologies into image-level, representation-level, and inference-level techniques depending on where they intervene within biometric recognition systems. Another taxonomy proposed by Cao et al. [30] organizes methods into pixel-level, representation-level, and semantic-level categories based on the level at which anonymization is applied during image generation or classification. While these taxonomies offer useful abstractions, they do not fully capture important dimensions such as target dependency, reversibility, perceptual artifacts, or support for multiple-image anonymization.

To address this gap, we propose a complementary, mechanism-centric classification of identity anonymization methods into seven categories, based on how identity suppression is achieved and the data processing pathway: k-Anonymity, Morphing, Generative Modeling, Reversible Face Anonymization, Adversarial Perturbations, Deepfakes, and Non-Generative Modeling. This enables systematic comparison along consistent axes including face generation type, perceptible artifacts, attributes affected, multiple-image support, target-identity dependency, and the presence of a user study (Table I).

A. *k-Anonymity*

Newton et al. [40] have shown that ad-hoc methods such as obfuscating eyes are not able to thwart facial recognition systems. Therefore, the authors have proposed a k-anonymity-based algorithm [41] termed as k-same that guarantees privacy-preservation with the factor $\frac{1}{k}$. Gross et al. [13] proposed a k-same select algorithm, which preserves the gender and expression in the identity anonymized images using a data utility function along with providing the guarantee of privacy-preservation. Better quality of identity anonymized images is produced using Model-based k-Same algorithm, which is based on Active Appearance Models [42] and mixtures of identity and non-identity components in face images. Jourabloo et al. [43] have proposed an attribute-preserving face de-identification technique that uses gradient descent to learn the weights for fusing k images. Cao et al. [32] introduced a novel framework that combines k-anonymity and differential privacy (DP) for face image anonymization. The framework obscures identity-revealing attributes in facial images while maintaining visual realism and utility, without relying on fixed target identities or full generative reconstruction.

B. *Morphing*

The techniques mentioned above, i.e., the k-same algorithm and their variants, replace original images with a set of identity anonymized images in an irreversible manner. Korshunov

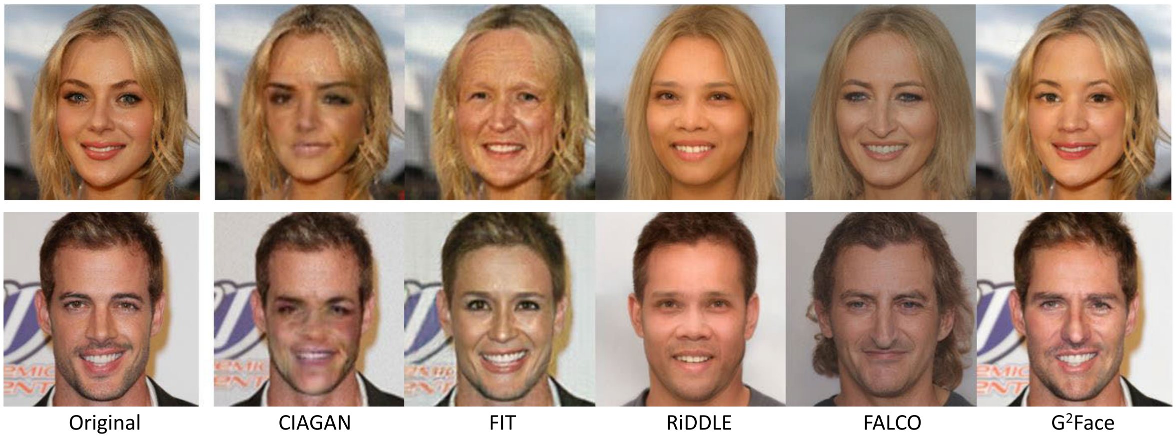

Fig. 2. Sample identity-anonymized images are taken from existing literature. The first column shows the original (non-anonymized) images, while the remaining columns show anonymized versions generated by different methods. Specifically, images in the second column are from CIAGAN [6]; the third column shows results from FIT [7]; the fourth from RiDDLE [8]; the fifth from FALCO [9]; and the last column from G^{2}Face [10].

TABLE I

COMPARISON OF EXISTING ALGORITHMS FOR IDENTITY ANONYMIZATION

Paper	Method	Category	Face generation type	Artifacts perceptible to humans	Attributes affected in face image	Multiple Image Anonymization	User Study	Target Identity based Anonymization
k-Same-Net [31]	K-Anonymity with Generative Deep Neural Networks	k-Anonymity	Visually dissimilar faces	Visible	Nose, Eye, Facial texture	No	No	Yes
Differentially Private k-Anonymity [32]	Differentially Private k-Anonymity with Attribute Editing	k-Anonymity	Visually dissimilar faces	Visible	Hair Color, Age, Facial texture	No	No	No
MIPGAN [33]	Morphing with GAN	Morphing	Visually dissimilar faces	Visible	Expression, skin tone	No	No	Yes
GAGM [34]	Synthesizes morphs with facial geometry and texture	Morphing	Visually dissimilar faces	Visible	Texture, Facial shape	No	No	Yes
LIVE [35]	Adversarial Autoencoder coupled with trained face classifier	Generative Modeling	Visually similar faces	Visible	Nose, Eye, Facial texture	No	Yes	Yes
CIAGAN [6]	Conditional GANs	Generative Modeling	Visually dissimilar and diverse set of images	Visible	Nose. Eyes, Makeup, Mouth region,	No	No	Yes
RIDDLE [8]	StyleGAN inversion encoder coupled with Latent Encryptor	Reversible Face Anonymization	Visually dissimilar and diverse set of images	Visible	Nose, Eyes, Mouth region	No	No	Yes
G2Face [10]	StyleGAN coupled with 3D geometric priors	Reversible Face Anonymization	Visually dissimilar and diverse set of images	Visible	Nose, Eyes, Mouth region	No	No	Dummy
TIP-IM [15]	Transferable Identity Perturbations via Iterative Masking	Adversarial Perturbation	Visually similar	Minimal	Minor edge noise	No	No	Yes
LowKey [14]	Leveraging Adversarial Attacks to Protect Social Media Users	Adversarial Perturbation	Visually similar	Minimal	Texture alterations	No	No	No
SimSwap [36]	ID Injection Module	Deepfake	Visually dissimilar and diverse set of images	Visible	Expression, Texture, Facial Shape	No	No	Yes
Disguise [37]	Face Swapping while ensuring differential privacy	Deepfake	Visually dissimilar and diverse set of images	Visible	Nose, Eyes, Mouth region	No	No	Synthetic
Attention Distraction [38]	Intrinsic and Extrinsic Attention Distraction	Non-generative Modeling	Visually dissimilar and diverse set of images	Visible	Nose, Eyes, Mouth region	No	Yes	Yes
PRO-Face C [39]	Client-side hybrid image obfuscation and complementary feature compensation	Non-generative Modeling	Visually dissimilar and diverse set of images	Visible	Nose, Eyes, Mouth region	No	No	No
Proposed+$\mathcal{L}_p$ (PrIdentity)	Adversarial perturbation masks identity via p-norm optimization	Adversarial Perturbations	Generate visually similar image	Not visible	No attributes are affected	Yes	Yes	No

and Ebrahimi [11] proposed a morphing-based method for visual privacy-preservation, where facial keypoints (e.g., eyes, nose) from source and target images are interpolated using a weighted sum of intensities to control privacy strength. The images are segmented via Delaunay triangulation, and the original image can be approximately recovered through unmorphing. In a related work [44], they applied image warping by perturbing facial keypoints with weighted random offsets to control warping strength, followed by computing a transformation and its inverse for approximate reconstruction. Both approaches were evaluated on the Yale dataset. Additionally, Korshunov and Ebrahimi [45] compared various privacy filters including blurring, pixelation, morphing, masking, and warping across PCA, LDA, and LBP based face recognition models, finding that morphing based filtering achieved the strongest privacy-preservation. Zhang et al. [46] presented MIPGAN, a GAN-based framework to generate high-quality facial morphs that are photorealistic and resistant to morphing detection, enabling strong face morphing attacks, which could be used for identity anonymization. Deng et al. [34] proposed a geometry-aware generative model (GAGM), which synthesizes morphs with both facial geometry and texture based on dual invertible networks. GAGM showed resistance to detection systems and can be leveraged for privacy-preservation.

C. Generative Modeling

Various techniques have been proposed for face de-identification via generation approaches [15], [47]. Meden et al. [48] used a generative network to synthesize artificial surrogate faces for anonymity while preserving the non-identity-related information. Wu et al. [49] proposed a Privacy-Protective-GAN framework for face de-identification that consists of two modules, the contrastive loss verificator and the structural similarity index (SSIM) regulator.

To improve the trade-off between privacy and data utility with improving the naturalness of the generated image through StyleGAN, a clustering-based scheme is proposed by Le et al. [50]. Gafni et al. [35] proposed an adversarial autoencoder network for real-time de-identification in the videos. The

TABLE II

COMPARISON OF EXISTING IDENTITY ANONYMIZATION METHODS ACROSS PRIVACY, DATA UTILITY, AND GENERALIZABILITY. THE PROPOSED PRIDENTITY IS THE ONLY APPROACH THAT SIMULTANEOUSLY ACHIEVES PRIVACY, DATA UTILITY, AND GENERALIZABILITY

Method	Privacy	Data Utility	Generalizability
k-Same-Net [31]	Moderate	Low	Low
Differentially Private k-Anonymity [32]	High	Low	Low
MIPGAN [33]	High	Low	Moderate
GAGM [34]	High	Moderate	Moderate
LIVE [35]	High	Low Moderate	Moderate
CIAGAN [6]	High	Low Moderate	Moderate
RIDDLE [8]	High	Moderate	Moderate
G2Face [10]	High	High	Moderate
TIP-IM [15]	High	High	Moderate
LowKey [14]	High	High	Moderate
SimSwap [36]	High	Moderate	Low
Disguise [37]	High	Moderate	Moderate
Attention Distraction [38]	Moderate	High	Moderate
PRO-Face C [39]	High	Low	High
Proposed + $\mathcal{L}_p$ (PrIdentity)	**High**	**High**	**High**

proposed network takes the source video as input and the generated de-identified output video has a similar-looking person. This network comes with a major restriction that it changes the skin-tone of the face in the output video and cannot preserve the original texture. Maximov et al. [6] proposed a Conditional Identity Anonymization Generative Adversarial Network (CIAGAN) for identity anonymization in both images and videos. In their proposed technique, trained GAN generates fake identities which are not present in the training set. Chen et al. [51] used DCGANs to generate a similar-looking face image to capture all vital face attributes. Since the images are generated, the by-product is that the identity is masked. A similar work from Lin et al. [52] focused on face de-identification through an improved version of UNet as the generator while using two discriminators. This architecture is termed FPGAN, which is trained using a weighted combination of pixel loss, content loss, and adversarial loss. The authors mathematically prove the convergence of the proposed architecture.

Shamshad et al. [53] proposed a two step method to search for adversarial latent codes and used StyleGAN to generate the anonymized image. The proposed method make changes in the makeup of the face to fool the models. Wen et al. [54] proposed a IDeudemon method which uses a divide and conquer strategy for face de-identification. The proposed method obfuscates the 3D disentangled ID code using NeRF model and further used face parsing maps and GAN for identity anonymization. This method changes the geometry of identity related features eyes, nose, ears, mouth, and facial bones. Lopez et al. [55] proposed a hardware level face anonymization, where the proposed method learns the optical encoder to generate the face heatmap and generates a new anonymized face image using a reference face image and face heatmap.

Although these methods are able to achieve good quantitative results for face de-identifications but these methods suffers from preserving the original facial attributes such as eye color, shape of nose, skin tone, and facial expression.

D. Reversible Face Anonymization

In the work from Wen et al. [56], an IdentityMask framework is proposed for reversible face de-identification in videos. This framework uses deep motion flow for avoiding per-frame evaluation. Kuang et al. [57] proposed two-stage latent representation reorganization (LReOrg) framework for privacy-preservation which uses bidirectional network optimized using keyword-based swap training strategy with a multi-task loss. This method changes the visual appearance of identity related features including nose, eyes, expressions etc. Li et al. [8] proposed RiDDLE for face de-identification. The RiDDLE uses encryption decryption based scheme to recover the original images. Zhang et al. [58] proposed another work FaceRSA, which is on similar lines of encryption decryption for face anonymization and recovery. Yang et al. [10] proposed a reversible face anonymization method termed as G^{2}Face. The proposed methods extracts geometric information from 3D face and use generative priors in a multi-scale manner for reversible face anonymization. This method changes identity related features such as nose, mouth shape, shape of eyebrows etc. These works are primarily inspired from GAN based methods identity anonymization. Fig. 2 presents sample identity anonymized images from prior work, offering a clear visual comparison of existing approaches; the figure also highlights their limitations and motivates the proposed method, i.e. the need to preserve non-identity features while maintaining high visual fidelity, and avoiding the reliance on target identities or generative reconstruction frameworks. Table I shows the comparison of existing algorithms with the proposed algorithm for identity anonymization.

E. Deepfakes

In the literature, researchers also leverages the methods used to generate deepfakes for identity anonymization. In the work from Zhu et al. [59], a deepfake based technique is used in medical domain to examine Parkinson's disease through movement in facial key points. The authors used faceswap for the de-identification of faces while keeping facial key points invariant. Cai et al. [37] proposed Disguise method for face de-identification which is based on a face swapping method. This method changes the visual appearance of identity related attributes including nose, eyes. expressions etc. Chen et al. [36] proposed SimSwap, a face-swapping framework that is efficient, identity-consistent, and expression-preserving. It aims to generalize to arbitrary identity pairs without requiring pair-specific training, unlike many prior face-swapping models.

F. Non-Generative Modeling

Kuang et al. [38] proposed a new method for face identity anonymization, which extracts the intrinsic identity information in feature space and extrinsic identity information such as visual cues in visual space. The proposed method changes the appearance of identity related attributes such as eye color, nose shape etc. To preserve privacy in face datasets, Singh et al. [60] proposed a two-step sanitizing framework for the privacy-preservation of the entire dataset. In the first step, a global decoupling process decomposes raw data into sensitive and non-sensitive latent representations. The next step consists of local sampling to synthetically generate sensitive information with differential privacy and merge it with non-sensitive latent features while preserving privacy. Jiang et al. [61] proposed the DartBlur method for privacy-preservation, which generates blurred faces using DNN. Yuan et al. [39] proposed a client server based framework PRO-Face C, which performs face recognition in a privacy-preserving manner. The proposed method obfuscates the image on the client side and uses a feature-compensation mechanism to optimize the equilibrium between privacy-preservation and utility maximization on the server side.

G. Adversarial Perturbations

In the literature, few methods used adversarial perturbations for identity anonymization. Li and Lin [47] proposed a four stage framework called AnonymousNet. The framework consists of facial attribute estimation, privacy-metric-oriented face obfuscation, directed natural image synthesis, and adversarial perturbation as the four stages. A targeted identity-protection iterative method (TIP-IM) via generating identity perturbation masks is proposed by Yang et al. [15]. For anonymizing the identities, TIP-IM generates the perturbation masks while ensuring the transferability of the generated identity anonymized image. The training and generation of masks is done in a white-box setting while the testing and evaluation is done in a black-box setting. Cherepanova et al. [14] proposed LowKey, an adversarial perturbation based method that protects user photos from being exploited by facial recognition systems, especially when shared on social media platforms. Zhong and Deng [62] used adversarial perturbations to generate person specific mask. However, the authors used 10 images to generate the privacy mask, which may not be available in real world scenarios. Liu et al. [63] address the optimization based problem for class specific face de-identification. The authors discovered the optimization problem in large and small batch optimization and proposed the Gradient Accumulation (GA) for gradient stability. No visual outputs of the anonymized images are shown. Yang et al. [64] proposed a face de-identification method, which is primarily applicable in cloud platforms only for storing images in secured manner. The proposed method transforms the images such that it is unrecognizable to humans and only recognizable to face recognition models.

Based on the existing literature of all seven categories, adversarial perturbation–based anonymization offers several key advantages compared to other methods. First, it can be target-free, avoiding the need for external identity images. Second, achieves high visual fidelity through carefully regularized, imperceptible perturbations; demonstrates strong cross-model transferability to unseen face recognition systems without re-training. Third, it scales efficiently by applying a single uniform perturbation across multiple images of the same individual. The proposed PrIdentity leverages these strengths while overcoming a key limitation of fixed $\mathcal{L}_1$ or $\mathcal{L}_2$ norms by learning the optimal $\mathcal{L}_p$ norm during training. This enables an adaptive trade-off between sparsity and smoothness, allowing PrIdentity to jointly satisfy the critical properties of privacy, visual fidelity, and generalizability, which prior approaches address only partially or in isolation. Table II shows the comparison of existing methods across privacy, data utility, and generalizability. High, Moderate, and Low categories are assigned based on empirical performance. The proposed PrIdentity achieves High in all three due to strong identity suppression, preserved visual quality, and robust cross-model transferability.

III. Privacy-Preserving Identity Anonymization

Existing approaches rely on target face images, which are fused with input face images for identity anonymization. This results in visually distorted images and loss of data utility. Other approaches modify the skin tone and change the nose structure and eye color to suppress identity information, which hinders the applicability for different applications. Therefore, in this research, we have proposed a novel privacy-preserving identity anonymization or PrIdentity algorithm that ensures privacy, data utility, and generalizability across different unseen face recognition models.

The proposed PrIdentity algorithm perturbs the input face image to suppress identity information and preserve the visual appearance of the perturbed image. In order to maintain a trade-off between data utility and privacy, we have used the $\mathcal{L}_p$ norm-based regularization technique for learning adversarial perturbation, where p in $\mathcal{L}_p$ norm is a learnable parameter. We further extended the proposed algorithm to anonymize multiple images simultaneously via learning a single uniform perturbation. Fig. 3 illustrates the pipeline of the proposed algorithm which can be interpreted as: "Learn a perturbation **P** for an input image **I** such that on adding perturbation **P** to the input image **I**, the output image **A** is identity anonymized while maintaining the balance between data utility and privacy". The detailed steps of the proposed algorithm are discussed below.

A. Problem Definition

Given an input face image $\mathbf{I}_k \in [0, 1]$ from a subject $\mathbf{S} = \{\mathbf{I_1}, \mathbf{I_2}, \ldots, \mathbf{I_m}\}$, we aim to generate an anonymized image $\mathbf{A}_k$ such that (i) identity information is suppressed, while (ii) the visual appearance of the image is preserved. We achieve this by learning a perturbation **P** that, when added to the original image, produces an anonymized version:

$$\mathbf{A}_k = \mathbf{I}_k + \mathbf{P}, \quad \text{such that} \quad C(\mathbf{A}_k) \neq C(\mathbf{I}_k) \tag{1}$$

where $C(.)$ is the function that outputs the identity of an image, and the anonymized image $\mathbf{A_k}$ is required to be classified as a different identity than the original image $\mathbf{I_k}$.

Let $\phi(\mathbf{W}, \mathbf{b})$ be any pre-trained face recognition model that takes an image as an input and outputs its identity representation vector. Thus, the identity representation vector corresponding to an input image $\mathbf{I}_k$ and anonymized image $\mathbf{A}_k$ is obtained as:

$$\mathbf{Z}_{\mathbf{I_k}} = \phi(\mathbf{W}\mathbf{I_k} + b) \text{ and } \mathbf{Z}_{\mathbf{A_k}} = \phi(\mathbf{W}\mathbf{A_k} + b), \tag{2}$$

where, $\mathbf{Z}_{\mathbf{I_k}}$ and $\mathbf{Z}_{\mathbf{A_k}}$ are the identity representation vector. **W** is the weight matrix and b is the bias of the model ϕ.

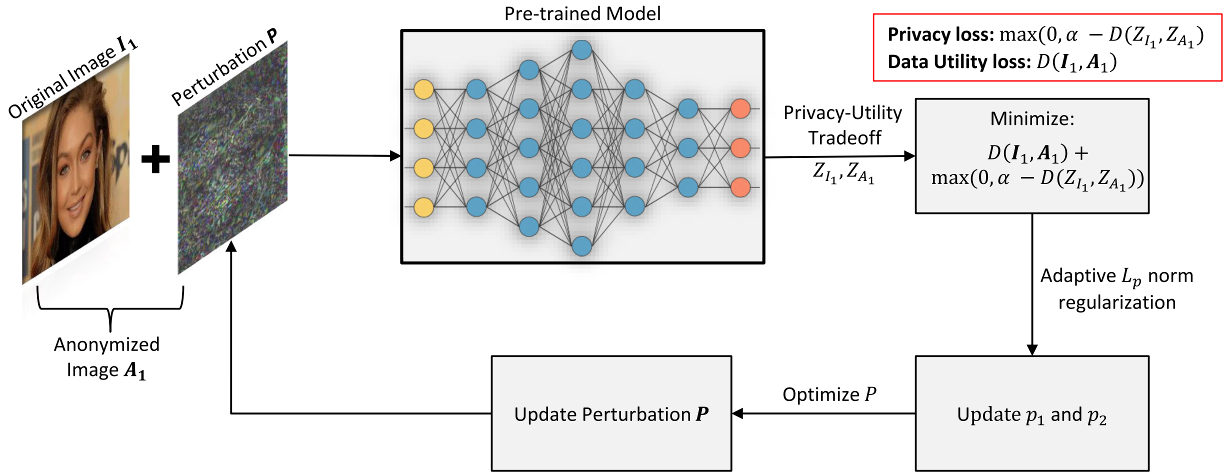

Fig. 3. Block diagram illustrating the pipeline of the proposed algorithm. The original image $\mathbf{I_1}$ is combined with a perturbation $\mathbf{P}$ to generate an anonymized image $\mathbf{A_1}$. A pre-trained model processes both the original and anonymized images to extract features Z_{I_1} and Z_{A_1}. The objective function minimizes data utility loss $D(\mathbf{I_1}, \mathbf{A_1})$ and privacy loss $\max(0, \alpha - D(Z_{I_1}, Z_{A_1}))$, guiding the optimization of perturbation $\mathbf{P}$ and norm parameters p_1 and p_2. The system iteratively updates $\mathbf{P}$, optimizing for both privacy protection and data utility.

B. Privacy-Data Utility Trade-off

The central challenge in identity anonymization is to suppress identity information (privacy) while preserving the visual quality of the image (utility). To achieve this balance, PrIdentity jointly optimizes two complementary objectives and unifies them through a margin-based formulation.

In order to anonymize the identity in an input image $\mathbf{I_k}$, the distance between $\mathbf{Z_{A_k}}$ and $\mathbf{Z_{I_k}}$ is maximized. Mathematically, it is written as:

$$\text{maximize } D(\mathbf{Z_{I_k}}, \mathbf{Z_{A_k}}), \tag{3}$$

where, $D(.)$ is the distance metric. A larger distance implies stronger identity suppression, thereby ensuring privacy. The above equation corresponds to the privacy measure of the input image $\mathbf{I}_k$ and satisfies the constraint of Equation 1.

In many real-world applications, it is important that the visual appearance of the image should not be distorted. Therefore, to preserve the visual appearance of an anonymized image, the distance between the anonymized image $\mathbf{A_k}$ and input image $\mathbf{I_k}$ is minimized.

$$\text{minimize } D(\mathbf{I_k}, \mathbf{A_k}), \tag{4}$$

where, Equation 4 corresponds to the data utility measure of the input image $\mathbf{I}_k$. It constrains the perturbation to be imperceptible, thereby maintaining high data utility. Using Equation 3 and 4, the objective function is written as:

$$\text{minimize } D(\mathbf{I_k}, \mathbf{A_k}) - D(\mathbf{Z_{I_k}}, \mathbf{Z_{A_k}}). \tag{5}$$

The perturbation $\mathbf{P}$ is learned by minimizing the above objective function. In order to control the separation among the identity representation $\mathbf{Z_{A_k}}$ and $\mathbf{Z_{I_k}}$, a margin parameter α is introduced. The margin parameter α ensures that the identity embeddings $\mathbf{Z_{A_k}}$ and $\mathbf{Z_{I_k}}$ should be at least α units, preventing trivial solutions where embeddings remain close. The objective function is thus updated to:

$$\text{minimize } D(\mathbf{I_k}, \mathbf{A_k}) + max(0, \alpha - D(\mathbf{Z_{I_k}}, \mathbf{Z_{A_k}})). \tag{6}$$

where, the first term promotes visual fidelity, while the second enforces privacy through a margin constraint in the embedding space. Together, they establish the fundamental privacy-utility trade-off in PrIdentity.

C. Adaptive L_p Norm Regularization

In many real-world applications, maintaining a good balance between data utility and privacy is of paramount factor. For this purpose, we have used the p-norm function as a distance metric, where p is a parameter to be learned. Thus, the objective function is updated as:

$$\text{minimize}\Bigg(\Big(\sum_{x,y}\|\mathbf{I_k}(x,y) - \mathbf{A_k}(x,y)\|^{p_2}\Big)^{\frac{1}{p_2}} + max\Bigg(0, \alpha - \Big(\sum_{i=1}^{d}\|\mathbf{Z_{I_{k,i}}} - \mathbf{Z_{A_{k,i}}}\|^{p_1}\Big)^{\frac{1}{p_1}}\Bigg)\Bigg), \tag{7}$$

where, p_1 and p_2 are the parameters corresponding to the privacy and data utility measure, respectively. In the literature, $\mathcal{L}_1$ norm where $p = 1$ and $\mathcal{L}_2$ norm where $p = 2$ are widely used in machine learning and deep learning applications. However, it is challenging to decide the suitable value of p-norm for a given application. To overcome this challenge, in this research, we have treated $p \in [1, 2]$ as a parameter and learned the optimal p to balance the privacy and data utility. To learn the value of p, we have used the SGD optimizer where we randomly initialize its value in the range [1, 2] and use the clip function to bound its value.

D. Generalizability

To ensure generalizability, it is important that the learned perturbation must be transferable to other deep models. In the literature, researchers have shown that the iterative learning of a perturbation generally overfits the surrogate model and hence performs poorly when evaluated on the target model. In general, the $\mathcal{L}_2$ norm is sensitive to outliers and is prone

TABLE III
DETAILS OF THE EXPERIMENTS PERFORMED TO EVALUATE THE PERFORMANCE OF THE PROPOSED ALGORITHM

Experiment	Dataset	Evaluation Type	Gallery Size	Description
Single Image Anonymization	CelebA	Face Identification	1, 2, 3, 4	This experiment is performed to show the performance with face identification. There is no literature available for the comparison.
	LFW	Face Verification	NA	This experiment is performed to showcase the performance in face verification setting. Evaluation on the LFW dataset in the verification setting is the standard protocol followed in the literature for the comparison.
Multiple Image Anonymization	CelebA	Face Identification	2, 4	This experiment is performed to showcase the performance in face identification setting. There is no literature available for the comparison.
Bounding Box Prediction	CelebA-HQ	Face Detection	NA	This experiment is performed to showcase the performance for face detection. Evaluation on the CelebA-HQ dataset for face detection is the standard protocol followed in the literature for comparison.

to overfitting. On the other hand, the $\mathcal{L}_1$ norm underfits the target model while generating adversarial perturbations. In our proposed algorithm, we address this problem by learning the optimal value of p in the range [1, 2], which results in robust perturbation that is transferable to unseen target face-recognition models. PrIdentity supports two use cases for identity anonymization:

1) Case I - Single Image Anonymization: This case corresponds to the scenario, where a user anonymizes a single image only. In this case, the perturbation will be different for each image for a given user and therefore, it provides more control to the user over the privacy of each image. For single image anonymization, we use $k = 1$ and Equation 6 is updated as:

$$\text{minimize } D(\mathbf{I_1}, \mathbf{A_1}) + max(0, \alpha - D(\mathbf{Z_{I_1}}, \mathbf{Z_{A_1}})), \quad (8)$$

where, first term is minimizing the distance between original and anonymized images while the second term is maximizing the distance between the face representations of the original and anonymized images.

2) Case II - Multiple Image Anonymization: This case corresponds to the scenario, where a user anonymizes multiple images of the same user. In this scenario, a single uniform perturbation $\mathbf{P}$ is learned for all the input images. Let n be the number of images to be anonymized. Thus, the objective function is written as:

$$\text{minimize} \sum_{k=1}^{n} \{D(\mathbf{I_k}, \mathbf{A_k}) + max(0, \alpha - D(\mathbf{Z_{I_k}}, \mathbf{Z_{A_k}}))\}. \quad (9)$$

Here, the above Equation iterated over the multiple images of the same user. In this case, the user has comparatively less control over the privacy as the perturbation will be the same for n images of a given user. However, this use case has relatively computationally efficient compared to single image anonymization.

To summarize, the PrIdentity algorithm integrates: (i) a perturbation generation module, (ii) a privacy loss, (iii) a utility loss, (iv) adaptive p-norm regularization, and (v) single vs multiple image anonymization. Together, these components enable a privacy-preserving face anonymization approach that balances identity suppression, data utility, and cross-model robustness.

IV. EXPERIMENTAL SETUP

In order to evaluate the performance of the proposed PrIdentity algorithm, two experiments are performed: 1) Single Image Anonymization and 2) Multiple Image Anonymization. The first experiment represents the scenario where only a single image for each identity is anonymized. In this experiment, images are anonymized by learning and adding a perturbation for each image. The second experiment represents the scenario where multiple images for each identity are anonymized simultaneously. In this experiment, a single uniform perturbation is learned to anonymize multiple images of the same identity. The comparison is performed with existing state-of-the-art algorithms. All the experiments are performed for $p = 1$, $p = 2$, and $1 < p < 2$ i.e $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_p$ norm, respectively. Here, $\mathcal{L}_p$ norm represents the case where the parameters p_1 and p_2 corresponding to privacy and data utility (visual appearance) are learned. Table III shows the details of the experiments. To quantify the data utility of the proposed algorithm, we have computed the bounding box distance on the CelebA-HQ dataset and compared the results with the existing algorithms.

To showcase the effectiveness of the proposed PrIdentity algorithm from the user's perspective, we have also conducted a user study on the original and their corresponding perturbed images. More details and results of the user study are discussed in Section V-D. Next, we discuss the data sets and protocol in detail. All the datasets used for evaluation in this research are publicly available to the research community.

LFW dataset [21] consists of 13,233 images of 5,749 subjects. To evaluate the performance of the proposed PrIdentity algorithm for face verification, standard view 2 of this dataset consisting of 6000 image pairs is used. One image from each pair is considered for identity anonymization.

CelebA dataset [22] contains 202,599 face images of more than 10k celebrities. The performance is evaluated for the face identification task. We have performed experiments on the test set, which consists of around 1k identities. The images of each subject are divided into the gallery and probe sets. Only gallery-set images are considered for identity anonymization. A single-image anonymization experiment is performed with a gallery set having 1, 2, 3, and 4 images corresponding to each identity, while a multiple-image anonymization experiment is performed with a gallery set having 2 and 4 images.

CelebA-HQ dataset [23] comprises 30,000 high-quality images. The images in this dataset are the high-quality version of the CelebA dataset. We used these images to generate the identity anonymized images using the proposed algorithm and further compared the face detection performance on original and identity anonymized images.

Face images from all three datasets are detected using Multi-Task Cascaded Convolutional Neural Networks (MTCNN) [65]. To measure the anonymized images' data utility, we have computed the Structural similarity index measure (SSIM) [66] score for each anonymized image corresponding to the given original input image. For measuring the data utility U of the complete dataset with anonymized images, we have taken the average of the SSIM score. The data utility score 1 represents that the image's visual appearance is completely preserved, while the data utility score 0 represents the image is completely distorted. A given dataset's privacy score Pr is measured as $Pr = (100 - \text{Accuracy})/100$. This means if the obtained accuracy is 100%, then the privacy score would be $\frac{100-100}{100}$, i.e., 0. On the other hand, if the obtained accuracy is 0%, then the privacy score would be $\frac{100-0}{100}$, i.e., 1. Ideally, both data utility U and privacy score Pr should be 1. To quantify the trade-off between data utility and privacy, we have taken the mean of data utility and privacy score. Mathematically, it is written as:

$$T = \frac{U + Pr}{2} \tag{10}$$

where, U and Pr represents the data-utility and privacy score, while T represents the trade-off score.

All images are anonymized corresponding to the ResNet50 [67] model pre-trained on the VGGFace2 [68] dataset for both the experiments. To evaluate the proposed algorithm's generalizability across different unseen face recognition models, we have reported the recognition performance using VGGFace [69], ArcFace [70], and LightCNN-29 [71]. The implementation details of the proposed algorithm are discussed in the following subsection.

A. Implementation Details

All the experiments are performed on a DGX station with Intel Xeon CPU, 256 GB RAM, and four 32 GB Nvidia V100 GPU cards. The implementation details for learning perturbation and evaluating performance on different face recognition models are given below.

Perturbation Learning: To learn a perturbation for both experiments, the proposed algorithm is implemented in Tensorflow 2.3.1. The perturbation is learned corresponding to the ResNet50 model pre-trained on the VGGFace2 dataset. The learning rate is set to 0.1 during training, and the perturbation is trained for 120 iterations. The margin α is set to 2 during the experiments.

For evaluating performance using the VGGFace model [69], the original and anonymized images are resized to 224 $\times$ 224. While evaluating performance using ArcFace [70] and LightCNN-29 [71] models, features are extracted after resizing the images to 128 $\times$ 128 resolution grayscale images. The distance between features of images is computed using cosine similarity.

B. Fair Comparison With Existing Methods

In the recent literature of face identity anonymization, it has been observed that the researchers are making unfair comparison with the existing methods to showcase state of the results. For example, ToonerGAN [72], which generates the caricature/cartoon images achieved state-of-the-art results on the LFW dataset by achieving 0.009 TPR at FPR of 0.001 and showed comparison with existing methods such as CIAGAN [6], which generates photo realistic anonymized images having some modifications in identity related attributes such as eye and nose. In order to make a fair comparison of privacy quantitatively, it is important that the use case or data utility of the existing methods should be similar. In this research, we have anonymized the face images for social media based applications, where the automated face recognition algorithms should not be able to recognize the anonymized image of a user uploaded on the social and the user perceives an anonymized image on social media as real image.

In order to make a fair comparison, we have compared the proposed algorithm with the state-of-the-art (sota) GAN based methods such as CIAGAN [6], RIDDLE [8], LIVE [35], TIP-IM [15], FIT [7], FALCO [9], and G2Face [10] for identity anonymization. Although, these methods modify the identity related attributes such as shape of nose, eye color, facial expression etc and are not perfectly suitable for the social media use case but these methods are the closest ones to compare. We have also compared the bounding box distance of the proposed algorithm against these methods to showcase the data utility of the proposed algorithm.

V. Performance Evaluation

The quantitative results of the proposed PrIdentity algorithm are evaluated for single and multiple-image anonymization. The qualitative and quantitative comparison is performed with the existing algorithms. For qualitative evaluation, the performance obtained through user study is also reported for a combination of both single and multiple image anonymization. The results for all three experiments are discussed in the following subsections.

A. Single Image Anonymization

In this experiment, a perturbation is learned corresponding to each image for the identity anonymization task. The results for both CelebA and LFW are discussed below.

1) Results on the CelebA Dataset: Table IV presents the identification accuracy on both original and identity anonymized images generated by the proposed algorithm using $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_p$ norms. The results are reported for gallery sets containing 1 to 4 images per identity and evaluated at ranks 1, 5, and 10 using VGGFace, ArcFace, and LightCNN-29 face recognition models. Here, rank-k indicates the accuracy when the correct identity appears within the model's top-k predictions. On evaluating the proposed PrIdentity algorithm in a

TABLE IV

IDENTIFICATION ACCURACY (%) FOR SINGLE IMAGE ANONYMIZATION EXPERIMENT ON THE ORIGINAL AND IDENTITY ANONYMIZED IMAGES OBTAINED USING THE PROPOSED ALGORITHM WITH $\mathcal{L}_1$, $\mathcal{L}_2$, AND $\mathcal{L}_p$ NORM ON THE CELEBA DATASET. TO SHOWCASE THE GENERALIZABILITY OF THE PROPOSED ALGORITHM, THE RESULTS ARE REPORTED ON VGGFACE, ARCFACE, AND LIGHTCNN-29 MODELS, RESPECTIVELY. A LOW ACCURACY INDICATES A HIGH PERFORMANCE (BETTER PRIVACY)

	Original			**Proposed + $\mathcal{L}_1$**			**Proposed + $\mathcal{L}_2$**			**Proposed + $\mathcal{L}_p$**		
	Rank 1	Rank 5	Rank 10	Rank 1	Rank 5	Rank 10	Rank 1	Rank 5	Rank 10	Rank 1	Rank 5	Rank 10
Gallery Size 1												
VGGFace	45.42	61.71	68.04	0.20	0.70	1.40	0.20	0.80	1.70	0.30	0.80	1.30
ArcFace	64.46	76.66	80.94	5.40	14.03	19.71	2.93	8.52	12.99	2.55	7.50	11.44
LightCNN	83.91	90.53	92.43	18.04	32.52	40.31	10.60	22.69	29.09	9.57	19.83	26.17
Gallery Size 2												
VGGFace	57.30	73.27	79.05	0.10	0.67	1.34	0.15	0.72	1.18	0.25	0.72	1.03
ArcFace	76.17	87.2	90.27	7.84	18.13	24.86	3.90	10.76	16.08	3.19	9.18	13.94
LightCNN	91.39	95.63	96.48	24.00	44.12	51.37	13.67	27.68	36.30	12.61	26.32	33.95
Gallery Size 3												
VGGFace	63.16	78.25	83.55	0.14	0.42	0.94	0.14	0.52	0.98	0.14	0.38	0.98
ArcFace	81.74	90.55	93.09	9.00	20.98	28.65	4.46	11.59	17.07	3.40	10.35	15.52
LightCNN	93.26	96.32	97.08	27.95	47.78	56.83	17.01	32.87	41.36	14.94	30.36	38.71
Gallery Size 4												
VGGFace	81.44	86.23	87.33	0.16	0.61	1.31	0.16	0.51	0.96	0.21	0.51	0.99
ArcFace	84.27	92.13	94.31	10.03	22.64	30.58	5.25	13.65	19.67	3.92	11.09	16.99
LightCNN	94.14	96.88	97.51	30.84	51.53	60.83	19.03	36.41	45.59	16.94	32.76	42.24

TABLE V

DATA UTILITY (SSIM) SCORES FOR SINGLE IMAGE ANONYMIZATION EXPERIMENT ON THE ORIGINAL AND CORRESPONDING IDENTITY ANONYMIZED IMAGES OBTAINED USING THE PROPOSED ALGORITHM WITH $\mathcal{L}_1$, $\mathcal{L}_2$, AND $\mathcal{L}_p$ NORM ON THE CELEBA DATASET. A HIGH SCORE INDICATES HIGH PERFORMANCE

Gallery Size	$\mathcal{L}_1$	$\mathcal{L}_2$	$\mathcal{L}_p$
1	0.9004	0.8629	0.84187
2	0.9001	0.8628	0.8422
3	0.9000	0.8628	0.8423
4	0.8999	0.8625	0.8414

black-box setting using VGGFace, ArcFace, and LightCNN-29 face recognition models, our results show that PrIdentity successfully anonymizes face images, significantly reducing identification accuracy across all models. The performance degradation was stronger for ArcFace and LightCNN-29 compared to VGGFace, likely because these models use grayscale images while perturbations are learned on color images. Overall, the results demonstrate that PrIdentity effectively preserves individual privacy in real-world scenarios.

When comparing $\mathcal{L}_1$, $\mathcal{L}_2$, and the learnable $\mathcal{L}_p$ norm, the proposed algorithm with $\mathcal{L}_p$ outperformed fixed norms in rank-1, rank-5, and rank-10 accuracy, particularly for ArcFace and LightCNN-29. The fixed $\mathcal{L}_1$ norm introduces sparsity but may leave identity cues intact, while the $\mathcal{L}_2$ norm is sensitive to outliers and leads to visible perturbations. In contrast, the learnable p parameter adapts between 1 and 2, balancing sparsity and smoothness to avoid overfitting, yielding better privacy and generalization.

We also studied the scenario where multiple images per identity are stored in the gallery set. As the gallery size increased from 1 to 4 images, identification accuracy on original images increased substantially (e.g., from 64.46% to 84.27% in ArcFace), while accuracy on anonymized images increased marginally by only 1.37%. This shows that our method scales well in real-world applications where multiple images per subject are available, without significantly compromising privacy. Lastly, we observed that LightCNN-29 generally achieved the best identification accuracy on both original and anonymized images, making it the most robust model against our perturbations. This highlights an important limitation and motivates future work to improve anonymization against particularly robust FR architectures.

2) Effectiveness of $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_p$ Norm: In order to measure the effectiveness of individual norms for the proposed algorithm, it is important to evaluate the privacy score, data-utility score, and the trade-off between them. As mentioned above, lower accuracy represents better privacy. It implies that the $\mathcal{L}_p$ norm achieves better privacy compared to $\mathcal{L}_1$ and $\mathcal{L}_2$ norms. Table V shows the data utility scores obtained by computing SSIM scores. A higher value indicates better data utility. It is observed that the $\mathcal{L}_1$ norm has better data utility, followed by the $\mathcal{L}_2$ norm and $\mathcal{L}_p$ norm. However, the effectiveness of the algorithm should be measured by the trade-off between privacy and data utility. Fig. 4 shows the comparison of the trade-off score of $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_p$ norms at ranks 1, 5, and 10 respectively, corresponding to the LightCNN-29 model for gallery set having 3 images for each identity. It is observed that the $\mathcal{L}_1$ norm is performing worst at ranks 1, 5, and 10. On the other hand, the $\mathcal{L}_p$ norm performs best at rank 5 and 10, while on rank 1, it shows comparable performance with the $\mathcal{L}_2$ norm. These quantitative results indicate that the $\mathcal{L}_p$ norm effectively maintains a good balance between privacy and data utility. This trade-off can also be qualitatively visualized in Fig. 5.

3) Results on the LFW Dataset and Comparison With Existing Algorithms: To evaluate the performance on the LFW dataset, one image from each pair is anonymized. The

TABLE VI

IDENTIFICATION ACCURACY (%) FOR MULTIPLE IMAGE ANONYMIZATION EXPERIMENT ON THE ORIGINAL AND IDENTITY ANONYMIZED IMAGES OBTAINED USING THE PROPOSED ALGORITHM WITH $\mathcal{L}_1$, $\mathcal{L}_2$, AND $\mathcal{L}_p$ NORM ON THE CELEBA DATASET. TO SHOWCASE THE GENERALIZABILITY OF THE PROPOSED ALGORITHM, THE RESULTS ARE REPORTED ON VGGFACE, ARCFACE, AND LIGHTCNN-29 MODELS RESPECTIVELY. A LOW ACCURACY INDICATES A HIGH PERFORMANCE (BETTER PRIVACY)

	Original			**Proposed + $\mathcal{L}_1$**			**Proposed + $\mathcal{L}_2$**			**Proposed + $\mathcal{L}_p$**		
	Rank 1	Rank 5	Rank 10	Rank 1	Rank 5	Rank 10	Rank 1	Rank 5	Rank 10	Rank 1	Rank 5	Rank 10
Gallery Size 2												
VGGFace	57.30	73.27	79.05	0.10	0.61	1.44	0.20	0.77	1.18	0.25	0.56	1.13
ArcFace	76.17	87.2	90.27	20.62	38.21	46.83	7.31	17.67	24.17	5.38	13.7	19.91
LightCNN	91.39	95.63	96.48	47.93	67.67	74.93	24.7	42.44	50.81	21.2	38.15	46.63
Gallery Size 4												
VGGFace	67.33	81.44	86.23	0.10	0.56	1.07	0.10	0.53	1.05	0.18	0.64	1.15
ArcFace	84.27	92.13	94.31	42.22	63.08	71.28	11.49	24.56	32.9	8.35	19.58	27.29
LightCNN	94.14	96.88	97.51	73.45	88.51	91.21	37.80	61.84	68.14	32.48	53.06	61.92

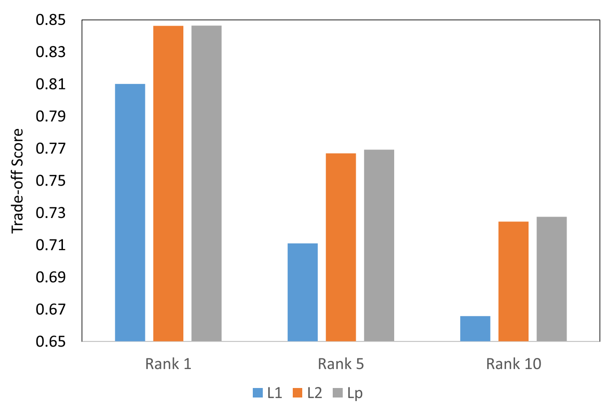

Fig. 4. Comparing trade-off score (T) for single image anonymization for gallery size 3 on the CelebA dataset.

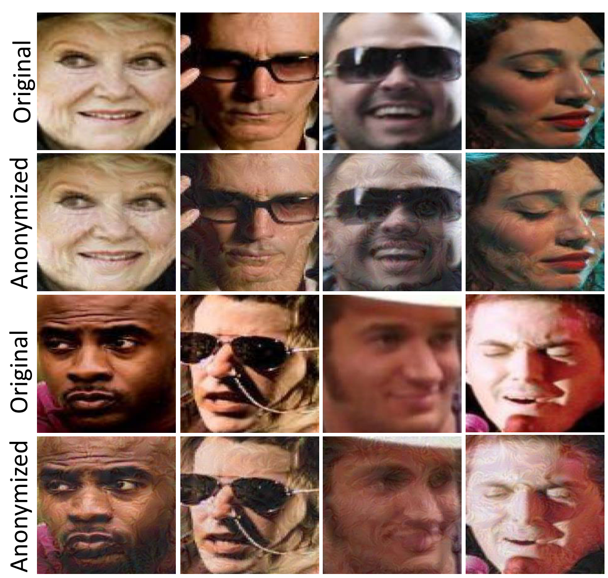

Fig. 5. Sample images before and after anonymization obtained using the proposed algorithm $\mathcal{L}_p$ norm (Best viewed in color).

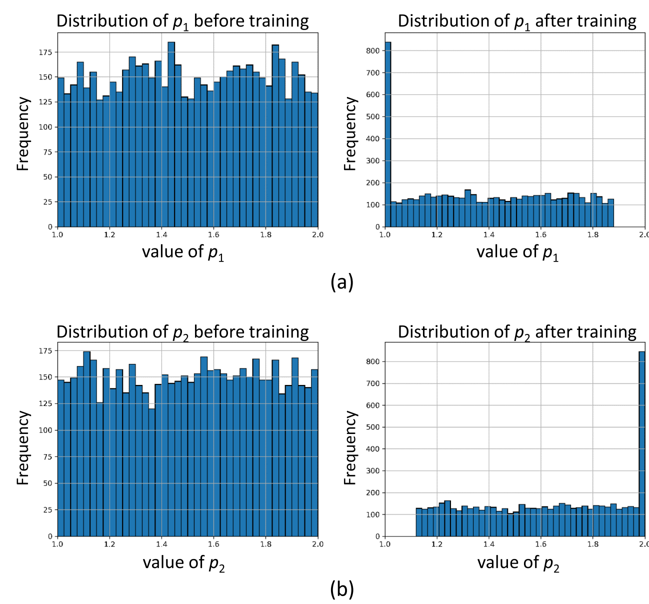

Fig. 6. Distribution of the learnable norm parameters p_1 (for privacy) and p_2 (for data utility) before and after training. **(a)** Histograms of p_1: initially sampled uniformly in [1.0, 2.0], the values converge toward 1.0 after training, favoring sparse, localized perturbations to effectively disrupt identity embeddings. **(b)** Histograms of p_2: also initialized uniformly, with post-training values concentrating near 2.0, promoting smooth perturbations that preserve visual fidelity.

TABLE VII

VERIFICATION PERFORMANCE OF THE PROPOSED ALGORITHM WITH $\mathcal{L}_1$, $\mathcal{L}_2$, AND $\mathcal{L}_p$ NORM AT FPR 0.001 ON THE LFW DATASET

	Original	$\mathcal{L}_1$	$\mathcal{L}_2$	$\mathcal{L}_p$
ArcFace	0.9940	0.0077	0.0047	0.002
LightCNN-29	0.9930	0.0127	0.0147	0.013

verification performance is evaluated on the anonymized images using the proposed algorithm with $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_p$ norms as shown in Table VII. We have observed that the proposed algorithm with $\mathcal{L}_p$ norm performs well and reduces the True Positive Rate to 0.013, corresponding to LightCNN-29 model. The better results on both face identification and verification tasks show the suitability of the proposed algorithm for different applications.

The results are also compared with state-of-the-art algorithms, as shown in Table VIII. FaceNet model pre-trained on the CASIA-Webface database and ArcFace model trained on MS1MV3 dataset are used for the comparison. Our proposed algorithm with $\mathcal{L}_p$ norm achieves the second-best result with

TABLE VIII
COMPARING THE TPR OF THE PROPOSED AND EXISTING IDENTITY ANONYMIZATION ALGORITHMS ON THE LFW DATASET AT 0.001 FPR

	FaceNet	**ArcFace**
Original	0.965	0.987
DIM [20]	-	0.042
MT-DIM [20]	-	0.040
LIVE [35]	0.035	0.025
CIAGAN [6]	0.019	0.021
FIT [7]	0.033	0.027
TIP-IM [15]	-	0.026
RIDDLE [8]	0.032	0.011
FALCO [9]	0.021	0.015
G2Face [10]	**0.009**	0.007
Proposed + $\mathcal{L}_p$	0.019	**0.002**

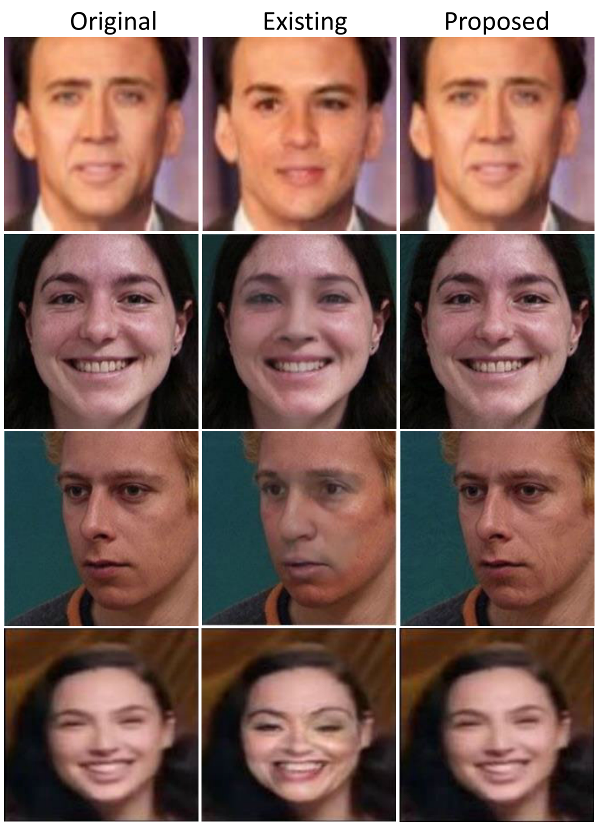

Fig. 7. Comparison of anonymized images generated using existing algorithms [6], [35] and the proposed algorithm for single image anonymization. Fourth row second column image is from Maximov et al. [6] while the other existing algorithm results are from Gafni et al. [35]. Images generated using existing algorithm are heavily distorted, whereas the proposed algorithm preserves the visual appearance.

0.019 True Positive Rate when evaluated using the FaceNet model pre-trained on CASIA-Webface dataset [73]. It achieved state-of-the-art (sota) results with 0.002 True Positive Rate when evaluated with ArcFace model. We would like to mention that the proposed algorithm does not require the target image for the anonymization while maintaining the visual appearance, which is not the case with other compared algorithms. Fig. 7 shows some anonymized sample images generated using existing algorithms and the proposed algorithms. It is clear from Fig. 7 that the visual distortions are present in the images generated using CIAGAN [6] and [35].

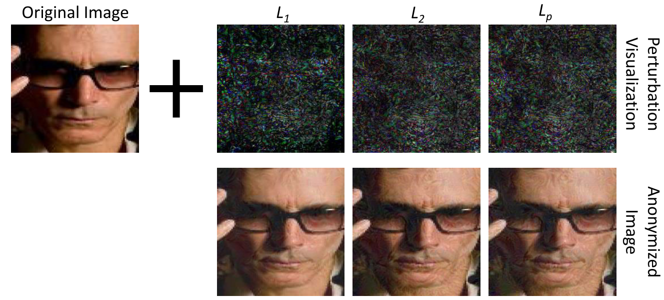

Fig. 8. Sample images before and after anonymization obtained using the proposed algorithm with $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_p$ norm on the CelebA dataset.

However, the visual appearance is preserved in the images generated using the proposed algorithm. CIAGAN uses an identity guidance discriminator, which requires target identity information as input to generate the anonymized image. As a result, the image generated using CIAGAN has some characteristics of target identity and thus visually distorts the image. The method proposed by [10] and [35] requires the target image or dummy embedding to generate the anonymized image. However, our proposed algorithm does not require any target image or dummy embedding and generates visually natural images.

4) Comparison With Perturbation Based Methods: Since adversarial perturbation–based methods are most directly related to our approach, we specifically compared PrIdentity against them to highlight its effectiveness. On the LFW dataset with ArcFace, TIP-IM [15], MT-DIM [20], and DIM [20] achieved TPR of 0.026, 0.040, and 0.042, respectively, showing that while these methods reduce identity leakage, a non-trivial amount of identity information remains detectable. While these methods successfully reduce identity leakage, a noticeable amount of identity information remains detectable. In contrast, PrIdentity lowers the TPR to 0.002, effectively eliminating recognizable identity traces. This constitutes more than an order-of-magnitude improvement over existing methods, underscoring the superior privacy-preservation capability of PrIdentity. Importantly, this strong suppression of identity cues is attained without compromising visual fidelity, as confirmed by both user studies and bounding box distance evaluations (discussed later).

5) Analysis of Learned p Distribution: To evaluate the stability and behavior of the learnable norm parameters p_1 (for privacy) and p_2 (for data utility), we present their distributions before and after training on the LFW dataset in Fig. 6. A detailed analysis is provided below:

Initial Distributions: Both p_1 and p_2 are initialized uniformly within the interval $[1.0, 2.0]$. As shown in the left column of Fig. 6, the distributions are relatively flat, confirming that the initialization is unbiased and does not favor any particular norm behavior.

Final Distributions: The right column of Fig. 6 shows the learned values of p_1 and p_2 after training. Specifically, p_1 (privacy) values predominantly converge near 1.0, indicating a preference for sparser, localized perturbations that effectively disrupt identity representations while preserving image quality.

In contrast, p_2 (data utility) values tend to cluster near 2.0, reflecting a preference for smoother, more distributed perturbations that better maintain low-level visual fidelity.

The optimization process remains well-behaved and bounded within the theoretical limits of [1.0, 2.0], with convergence patterns validating the robustness and stability of the proposed learning algorithm. These results reinforce the conceptual design of our approach: letting the model adapt the regularization behavior of adversarial perturbations through adaptive p-norms. This flexibility enables a better trade-off between privacy and utility than static norms.

B. Multiple Image Anonymization

This experiment is performed to showcase that the proposed algorithm can be extended to suppress multiple images simultaneously. In this experiment, multiple images of the same identity are anonymized via learning a single uniform perturbation. The experiments are performed by anonymizing 2 and 4 images of the same identity. Table VI shows the identification accuracy on the original and identity anonymized images obtained using the proposed algorithm with $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_p$ norms. Similar to the results of single image anonymization, the proposed algorithm with $\mathcal{L}_p$ norm is performing better when 2 and 4 images for each identity are anonymized in the gallery set via single uniform perturbation. The identification accuracy obtained for gallery set having 2 and 4 images corresponding to the VGGFace model at rank 10 is 79.05% and 86.23%, respectively, while after anonymization, the performance of the proposed algorithm with $\mathcal{L}_p$ norm is 1.13% and 1.15%. This shows the efficacy of the proposed algorithm.

It is important to note that the performance obtained using ArcFace and LightCNN-29 models are relatively lower compared to VGGFace model because the former models use grayscale images as an input while the perturbation is learned corresponding to color images. On comparing these results with a single image anonymization experiment for a gallery set having 2 and 4 images, it is observed that anonymizing a single image gives better results. However, anonymizing multiple images simultaneously is computationally efficient compared to single image anonymization. Overall, it's a trade-off between privacy and computation efficiency when comparing the performance of single and multiple image anonymization results. Fig. 8 shows some images obtained using the proposed algorithm with $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_p$ norm. It is clear from Fig. 8 that the visual appearance of the images is preserved after anonymization.

1) Comparison With OPOM [62]: The method proposed by OPOM [62] generates a person specific perturbation for identity anonymization. However, it has several limitations. OPOM uses 10 images of a person for identity anonymization, which is large and may not be available in real world scenario. However, the proposed algorithm requires 2-4 images for identity anonymization. OPOM was evaluated on 500 identities and achieved 13.5% rank 1 and 19.9% rank 5 identification accuracy given the gallery has 1 image only. On the other hand, the proposed algorithm has done more comprehensive evaluation with 1,000 identities with multiple images in gallery and achieved lower rank 1 identification accuracy 5.38% and rank 5 identification accuracy 13.7% identification accuracy for 2 images in gallery. The proposed algorithm also outperforms with gallery size of 4 images, which showcase that the proposed algorithm is more effective compared to existing algorithms for multiple image anonymization.

C. Bounding Box Prediction

In the previous sections, we showcase the performance of the proposed algorithm for identity anonymization. To further showcase that the proposed algorithm can be extended for other tasks, we evaluated the proposed algorithm for bounding box prediction in face images. Correct bounding box prediction helps in localizing the face, which is a prerequisite for face recognition tasks in general. We performed the bounding box distance prediction on the CelebA-HQ dataset as this is the standard dataset used in the literature for comparison. The bounding box distance is performed using MTCNN [65] on both original and identity anonymized images, respectively. The identity anonymized images are generated using $\mathcal{L}_p$ norm. Table IX shows the comparison of the results for bounding box distance using MTCNN. We achieved state-of-the-art (SOTA) results for the proposed algorithm, where the Euclidean distance is 2.65. The second best results are reported by RIDDLE [8], where the distance is 1.17 higher than the proposed algorithm. These results demonstrate the effectiveness of the proposed algorithm for preserving data utility while anonymizing images.

D. User Study

To evaluate the effectiveness of the proposed algorithm, we have performed a user study with 10 subjects (ages varying between 21 to 60 years, 4 males and 6 females) on 50 randomly chosen pairs of original and their corresponding perturbed images. These pairs consisted of both familiar and unfamiliar identities. The users were asked to perform two different tasks: 1) identify the original and perturbed image (generated through the PrIdentity algorithm) among the pairs and 2) identify if the original and their corresponding perturbed images belong to the same identity or not. Here, the first task measures the effectiveness of the proposed algorithm for data utility, while the second task measures the effectiveness for privacy.

For the first task, users could identify the original/perturbed images with an accuracy of 50%; interestingly, the errors are observed for both the classes, implying that the perturbations are not perceptible and the users are making a random guess. For the second task, the users were able to correctly identify all the pairs with a 100% matching rate since visual appearance is preserved after adding the learnt perturbations. It is important to note that the objective of the proposed algorithm is not to fool the human but to fool the face recognition models for privacy-preservation. In several real-world applications like social media, it is important that the anonymized images are identifiable by humans. The results indicate that the proposed algorithm successfully incorporates privacy as well as data utility.

TABLE IX
COMPARING THE PROPOSED ALGORITHM WITH STATE-OF-THE-ART IDENTITY ANONYMIZATION ALGORITHMS ON THE CELEBA-HQ DATASET FOR BOUNDING BOX DISTANCE PREDICTION USING MTCNN

Anonymization	Bounding Box Distance
DeepPrivacy [74]	4.65
CIAGAN [6]	20.38
FIT [7]	7.87
RIDDLE [8]	3.82
FALCO [9]	7.88
Diff-Privacy [75]	5.83
Proposed + $\mathcal{L}_p$	**2.65**

VI. CONCLUSION

This research presented a novel privacy-preserving identity anonymization (PrIdentity) algorithm that suppresses identity information by learning and adding a perturbation using $\mathcal{L}_p$ norm in the given image. To showcase the effectiveness of our proposed algorithm, single and multiple image anonymization experiments are performed on the popular LFW, CelebA, CelebA-HQ datasets. It is observed that the proposed PrIdentity algorithm maintains a good balance between privacy and data utility. Empirical results show that, with an appropriate p, the proposed algorithm can successfully anonymize the identity of the given image and generalize well across unseen target deep models.

In the future, the proposed PrIdentity algorithm will be extended to a weighted scheme between privacy and data utility. This scheme will be highly beneficial for different organizations that outweigh one over the other. For instance, an organization would prefer data utility over privacy; however, a law-enforcement agency would opt for privacy. The proposed algorithm can be further extended for a dual objective, where the algorithm will be responsible for anonymizing identities as well as facial attributes.

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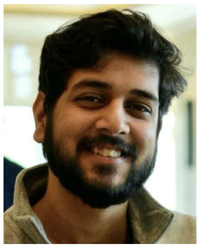
Saheb Chhabra received the Bachelor of Technology degree in electronics and communication from Lovely Professional University, Punjab, India, in 2013. Currently, he is pursuing the Ph.D. degree in computer science with IIIT-Delhi, India. His research interests include machine learning and deep learning with applications in privacy preservation. He has served as a Program Committee Member for AAAI. He has received the Best Student Paper Award at IEEE BTAS 2019.



Kartik Thakral (Student Member, IEEE) received the Bachelor of Technology degree in computer science from the College of Engineering Roorkee, India, in 2019. Currently, he is pursuing the Ph.D. degree with IIT Jodhpur. He has published multiple papers in top-tier venues, such as CVPR, ICLR, ICML, ECCV, and IEEE TRANSACTIONS ON BIOMETRICS, BEHAVIOR, AND IDENTITY SCIENCE. His research interests include computer vision, deep learning, and biometrics. He was awarded the Prime Minister Research Fellowship in October 2021.

Richa Singh (Fellow, IEEE) is currently a Professor with IIT Jodhpur, India. She has participated in several initiatives, including UIDAI (Aadhaar) and designing biometrics standards for e-government applications. She is a fellow of IAPR, INAE, and NASI. She was a recipient of the Kusum and Mohandas Pai Faculty Research Fellowship at IIIT-Delhi, the FAST Award by the Department of Science and Technology, India, and several best paper and best poster awards in international conferences. She has served as the Program Co-Chair for CVPR 2022, ICMI 2022, IJCB 2020, FG 2019, and BTAS 2016, and the General Co-Chair for FG 2021 and ISBA 2017. She is the Associate Editor-in-Chief of *Pattern Recognition* and the Co-EiC of *ACM AI Letters*. She was also the Vice President (Publications) of the IEEE Biometrics Council.

Mayank Vatsa (Fellow, IEEE) is currently a Professor at IIT Jodhpur, India. He has participated in several Indian government initiatives, including UIDAI (Aadhaar), designing biometrics standards for e-government applications, responsible AI, DigiYatra, and formation of TIH of computer vision and ARVR at IIT Jodhpur. He is a fellow of IAPR. He was a recipient of the prestigious Swarnajayanti Fellowship Award from the Government of India, the IEEE Meritorious Service Award, the A. R. Krishnaswamy Faculty Research Fellowship at IIIT-Delhi, and several best paper and best poster awards at international conferences. He has served as an Area/Associate Editor for *Pattern Recognition* and *Information Fusion*, the General Co-Chair for IJCB 2020, and the PC Co-Chair for FG 2021, AVSS 2021, IJCB 2014, and ICB 2013. From 2015 to 2018, he served as the Vice President (Publications) for the IEEE Biometrics Council, where he led the efforts to start IEEE TRANSACTIONS ON BIOMETRICS, BEHAVIOR AND IDENTITY SCIENCE.